

Notes on

Algebraic Geometry

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These are my very dense and somewhat disorganized notes on algebraic geometry, mostly from *The Rising Sea* by Vakil, with some supplements from the *Stacks Project* and Hartshorne's *Algebraic Geometry*.

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1 Sheaves

We begin with an example:

Example. The sheaf of smooth (C^∞) functions on a topological space $X = \mathbb{R}^n$ (or more generally on a manifold X) is the data of all smooth functions on all open subsets of X . On each open set $U \subset X$ we have a ring of smooth functions $\mathcal{O}(U)$.

We can restrict a smooth function given on an open set U to an open set $V \subset U$ via the restriction map $res_{V,U} : \mathcal{O}(U) \rightarrow \mathcal{O}(V)$. Note $res_{W,U} = res_{V,U} \circ res_{W,V}$.

If $\{U_i\}_{i \in I}$ is an open cover of open set U , and $f_1, f_2 \in \mathcal{O}(U)$, and $res_{U_i,U} f_1 = res_{U_i,U} f_2$, then $f_1 = f_2$.

Given $f_i \in \mathcal{O}(U_i)$ for all i s.t. $res_{U_i,U_i \cap U_j} f_i = res_{U_i,U_i \cap U_j} f_j$ for all i and j , then there is some $f \in \mathcal{O}(U)$ such that $res_{U_i,U} f = f_i$ for all i .

The germ of a smooth function f at a smooth point $p \in X$ is an object of the form $(f, \text{open set } U)$ such that $p \in U$, $f \in \mathcal{O}(U)$ modulo the relation that $(f, U) \equiv (g, V)$ if there is some open set $W \subset U$, $W \subset V$, $p \in W$ where $f|_W = g|_W$. In other words two functions that are the same in an open neighborhood of p (but may differ elsewhere) have the same germ.

The set of germs at p is the stalk $\mathcal{O}_p$, which is a (local) ring; if f is defined on U and g on V , $f + g$ is defined on $U \cap V$. Consider those germs vanishing at p which we denote $m_p \subseteq \mathcal{O}_p$. They form a (and in fact, the only) maximal ideal of $\mathcal{O}_p$.

Note: for a germ $[f]_p \in m_p$, define $\phi : m_p \rightarrow T_p^*X$ by $[f]_p \mapsto (df)_p$ where df is the differential of f at p . By the product rule (calculus always comes in handy!) $d(fg)_p = f(p)dg_p + g(p)df_p = 0$ if $f(p) = g(p) = 0$, so $m_p^2 \subseteq \ker \phi$. Hence ϕ descends to a linear map $\tilde{\phi} : m_p/m_p^2 \rightarrow T_p^*X$. Choose a local chart $(U; x^1, \dots, x^n)$ around p with $x^i(p) = 0$. Then each $x^i \in m_p$ and $([x^i]_p) = (dx^i)_p$. The covectors $\{(dx^i)_p\}_{i=1}^n$ form a basis of T_p^*X , so $\tilde{\phi}$ is surjective. In these same local coordinates, now choose $[f]_p \in m_p$ such that $df_p = 0$. A cute first-order Taylor's theorem gives $f(x) = \sum_{i=1}^n x^i g_i(x)$ for smooth g_i ; the vanishing differential implies $g_i(p) = 0$ for all i , hence each $g_i \in m_p$, therefore $[f]_p = \sum_{i=1}^n [x^i]_p [g_i]_p \in m_p^2$. Thus $\ker \phi = m_p^2$ so $\tilde{\phi}$ is injective. Thus the descended map is a linear isomorphism, natural in X and p . Hence the cotangent space at p can be identified with m_p/m_p^2 .

We define sheaves of sets; in the definition the category *Sets* can be replaced by any reasonable category. A *presheaf* of sets $\mathcal{F}$ on a topological space X is the following data:

- To each open $U \subset X$ we have a set $\mathcal{F}(U)$, whose elements are called *sections* of $\mathcal{F}$ over U . If the U is omitted, we implicitly refer to sections of $\mathcal{F}$ over X , called global sections.
- For each inclusion $U \hookrightarrow V$ of open sets we have a *restriction map* $res_{V,U} : \mathcal{F}(V) \rightarrow \mathcal{F}(U)$. If $f \in \mathcal{F}(V)$ we often write $f|_U := res_{V,U}(f)$, satisfying:
 - The map $res_{U,U}$ is the identity.
 - If $U \subset V \subset W$, $res_{W,U} = res_{V,U} \circ res_{W,V}$.

The *stalk* of a presheaf $\mathcal{F}$ at $p \in X$ is the set of germs $\mathcal{F}_p$ of $\mathcal{F}$ at p . Germs correspond to sections over some open set containing p , and two of these sections are considered the same if they agree on some smaller open set containing p . A useful equivalent definition of a stalk is

$$\mathcal{F}_p = \text{colim} \mathcal{F}(U)$$

The index category is a filtered set.

If $p \in U$ and $f \in \mathcal{F}(U)$ then the image of f in $\mathcal{F}_p$ is called the *germ* of f at p .

A presheaf is a *sheaf* if it satisfies the following axioms.

- *Identity*: For any open set U , if $\{U_i\}_{i \in I}$ is an open cover of U and $f_1, f_2 \in \mathcal{F}(U)$ and $f_1|_{U_i} = f_2|_{U_i}$ for all i , then $f_1 = f_2$.
- *Gluability*: If $\{U_i\}_{i \in I}$ is an open cover of U , then given $f_i \in \mathcal{F}(U_i)$ for all i such that $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j then there is some $f \in \mathcal{F}(U)$ such that $\text{res}_{U, U_i} f = f_i$ for all i .

Define the $\mathcal{F}|_U$ the restriction of $\mathcal{F}$ a sheaf on X to the open subset U to be the collection $\mathcal{F}|_U(V) = \mathcal{F}(V)$ for all open subsets $V \subset U$; this is a sheaf on U .

If U is an open subset of Y , $i : U \rightarrow Y$ is the inclusion, and $\mathcal{G}$ a sheaf on Y , then $i^{-1}\mathcal{G} \cong \mathcal{G}|_U$.

Example. Suppose X is a topological space, with $p \in X$, and S a set. Let $i_p : p \rightarrow X$ be the inclusion. Then $i_{p,*}S$ defined by

$$i_{p,*}S(U) = \begin{cases} S & \text{if } p \in U \\ \{e\} & \text{if } p \notin U \end{cases}$$

forms a sheaf, called the *skyscraper sheaf* supported at p . Here $\{e\}$ is any one-element set. There is an analogous definition for sheaves of abelian groups, except $i_{p,*}S(U) = \{0\}$ if $p \notin U$; and for sheaves with values in a category more generally, $i_{p,*}S(U)$ should be a final object.

Define $\underline{S}_{\text{pre}}(U) = S$ for all open sets U , forming the *constant presheaf* associated to S , with restriction maps the identity. If $\mathcal{F}(U)$ is the set of maps $U \rightarrow S$ that are locally constant, i.e., for any $p \in U$ $\exists$ local neighborhood of p where the function is constant, then this is a sheaf, called the *constant sheaf*, denoted $\underline{S}$.

Continuous maps to a topological space Y form a sheaf of sets on X . Suppose $\pi : X \rightarrow Y$ is a continuous map, and $\mathcal{F}$ is a presheaf on X . Then define a presheaf $\pi_*\mathcal{F}$ on Y by $\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}(V))$ where $V \subseteq Y$ is open. This is a presheaf on Y , and a sheaf if $\mathcal{F}$ is; it is called the *pushforward* or *direct image* of $\mathcal{F}$.

Suppose $\mathcal{O}_X$ is a sheaf of rings on a topological space X , i.e., a sheaf on X with values in the category of *Rings*. Then $(X, \mathcal{O}_X)$ is called a *ringed space*. The sheaf of rings, $\mathcal{O}_X$, is called the *structure sheaf* of the ringed space; sections of the structure sheaf over an open subset U are called *functions* on U ; functions on X are called *global functions*.

Just as we have modules over a ring, we have $\mathcal{O}_X$ -modules over a sheaf of rings $\mathcal{O}_X$. An $\mathcal{O}_X$ -module (or often $\mathcal{O}$ -module) is a sheaf of abelian groups $\mathcal{F}$ with the following additional structure. For each U , $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module. Furthermore, this structure should behave well with respect to restriction maps: if $U \subset V$, then the following commutes:

$$\begin{array}{ccc} \mathcal{O}_X(V) \times \mathcal{F}(V) & \xrightarrow{\text{action}} & \mathcal{F}(V) \\ \text{res}_{V,U} \times \text{res}_{V,U} \downarrow & & \downarrow \text{res}_{V,U} \\ \mathcal{O}_X(U) \times \mathcal{F}(U) & \xrightarrow{\text{action}} & \mathcal{F}(U) \end{array}$$

A *morphism of presheaves* of sets (or indeed of presheaves with values in any category) on X , $\phi : \mathcal{F} \rightarrow \mathcal{G}$, is the data of maps $\phi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ for all U behaving well with respect to restriction: if $U \hookrightarrow V$ then the following commutes.

$$\begin{array}{ccc}
\mathcal{F}(V) & \xrightarrow{\phi(V)} & \mathcal{G}(V) \\
\text{res}_{V,U} \downarrow & & \downarrow \text{res}_{V,U} \\
\mathcal{F}(U) & \xrightarrow{\phi(U)} & \mathcal{G}(U)
\end{array}$$

Morphisms of sheaves are defined identically: the morphisms from a sheaf $\mathcal{F}$ to a sheaf $\mathcal{G}$ are precisely the morphisms from $\mathcal{F}$ to $\mathcal{G}$ as presheaves. (The category of sheaves on X is a full subcategory of the category of presheaves on X). The morphisms of $\mathcal{O}_X$ -modules of a ring space $(X, \mathcal{O}_X)$ have the obvious definition.

Define an *isomorphism of ringed spaces* $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ as (i) a homeomorphism $\pi : X \rightarrow Y$ and (ii) an isomorphism of sheaves $\mathcal{O}_X$ and $\mathcal{O}_Y$ considered to be on the same space via π .

Let Sets_X , Ab_X , etc. denote the category of sheaves of sets, abelian groups, etc. on a topological space X ; let $\text{Mod}_{\mathcal{O}_X}$ denote the category of $\mathcal{O}_X$ -modules on a ringed space $(X, \mathcal{O}_X)$; let $\text{Sets}_X^{\text{pre}}$, etc denote the category of presheaves of sets, etc., on X .

Taking the stalk at $p \in X$ induces the *stalkification functor* $\text{Sets}_X \rightarrow \text{Sets}$, because the morphism of presheaves on X $\phi : \mathcal{F} \rightarrow \mathcal{G}$ induces a morphism of stalks $\phi_p : \mathcal{F}_p \rightarrow \mathcal{G}_p \forall p \in X$. The pushforward for a continuous map of topological spaces $\pi : X \rightarrow Y$ gives a functor $\pi_* : \text{Sets}_X \rightarrow \text{Sets}_Y$ where Sets can be replaced by other categories.

Suppose $\mathcal{F}, \mathcal{G}$ are two sheaves of sets on X (it suffices that $\mathcal{F}$ is a presheaf). Let $\mathcal{H}\mathcal{I}\mathcal{I}(\mathcal{F}, \mathcal{G})$ be the collection of data $\mathcal{H}\mathcal{I}\mathcal{I}(\mathcal{F}, \mathcal{G})(U) := \text{Mor}(\mathcal{F}_U, \mathcal{G}_U)$. This is a sheaf of sets on X , called "sheaf $\mathcal{H}\mathcal{I}\mathcal{I}$ ". Like Hom , $\mathcal{H}\mathcal{I}\mathcal{I}$ is a contravariant functor in its first argument and a covariant functor in its second argument. It does not commute with taking stalks, i.e. $\mathcal{H}\mathcal{I}\mathcal{I}(\mathcal{F}, \mathcal{G})_p \not\cong \text{Hom}(\mathcal{F}_p, \mathcal{G}_p)$.

If $\mathcal{F}, \mathcal{G}$ are sheaves of abelian groups on X , then $\mathcal{H}\mathcal{I}\mathcal{I}_{\text{Ab}_X}(\mathcal{F}, \mathcal{G})$ is defined by taking $\mathcal{H}\mathcal{I}\mathcal{I}_{\text{Ab}_X}(\mathcal{F}, \mathcal{G})(U)$ to be the maps as sheaves of abelian groups $\mathcal{F}|_U \rightarrow \mathcal{G}|_U$. If $\mathcal{F}$ and $\mathcal{G}$ are $\mathcal{O}_X$ -modules we define $\mathcal{H}\mathcal{I}\mathcal{I}_{\text{Mod}_{\mathcal{O}_X}}(\mathcal{F}, \mathcal{G})$ in an analogous way. We call $\mathcal{H}\mathcal{I}\mathcal{I}_{\text{Mod}_{\mathcal{O}_X}}(\mathcal{F}, \mathcal{O}_X)$ the dual of the $\mathcal{O}_X$ -module $\mathcal{F}$ and denote it $\mathcal{F}^\vee$.

Presheaves of abelian groups form an abelian category.

If $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of presheaves, the *presheaf kernel* $\ker_{\text{pre}} \phi$ is defined $(\ker_{\text{pre}} \phi)(U) := \ker \phi(U)$. It is a presheaf. The *presheaf cokernel* is defined similarly.

We call any property determined at the level of stalks *stalk-local*. This isn't true for presheaves and reflects the local nature of sheaves.

A section of sheaf of sets is determined by its germs, i.e. the natural map $\mathcal{F}(U) \rightarrow \prod_{p \in U} \mathcal{F}_p$ is injective.

An element $(s_p)_{p \in U} \in \prod_{p \in U} \mathcal{F}_p$ is said to consist of *compatible germs* if, $\forall p \in U$, $\exists$ some representative $(\tilde{s}_p \in \mathcal{F}(U_p), U_p \text{ open in } U)$ for s_p where $p \in U_p \subset U$ such that the germ of $\tilde{s}_p$ at all $q \in U_p$ is s_q . Equivalently there is an open cover $\{U_i\}$ of U and sections $f_i \in \mathcal{F}(U_i)$ s.t. if $p \in U_i$ then s_p is the germ of f_i at p . Clearly any section s of $\mathcal{F}$ over U gives a choice of compatible germs for U . Any choice of compatible germs for a sheaf of sets $\mathcal{F}$ over U is the image of a section $\mathcal{F}$ over U . Morphisms are moreover determined by stalks, i.e., if ϕ_1, ϕ_2 are morphisms from a presheaf of sets $\mathcal{F}$ to a sheaf of sets $\mathcal{G}$ that induce each stalk, then $\phi_1 = \phi_2$.

If $\mathcal{F}$ is a presheaf on X , then a morphism of presheaves $sh : \mathcal{F} \rightarrow \mathcal{F}^{sh}$ on X is a *sheafification* of $\mathcal{F}$ if $\mathcal{F}^{sh}$ is a sheaf, and for any sheaf $\mathcal{G}$ and any presheaf morphism $g : \mathcal{F} \rightarrow \mathcal{G}$ there exists a unique morphism of sheaves $f : \mathcal{F}^{sh} \rightarrow \mathcal{G}$ making the following commute:

$$\begin{array}{ccc}
\mathcal{F} & \xrightarrow{sh} & \mathcal{F}^{sh} \\
& \searrow g & \downarrow f \\
& & \mathcal{G}
\end{array}$$

This is a functor from presheaves on X to sheaves on X , unique up to a unique isomorphism assuming it exists. We can construct $\mathcal{F}^{sh}$ by defining $\mathcal{F}^{sh}(U)$ as the set of "compatible (families of) germs" of the presheaf $\mathcal{F}$ over U , i.e.

$$\mathcal{F}^{sh} := \{(f_p \in \mathcal{F})_{p \in U} : \forall p \in U, \exists \text{ open neighborhood of } p, V \subset U, \text{ and } s \in \mathcal{F}(V) \text{ with } s_q = f_q \forall q \in V\}$$

where s_q means the germ of s at q ; the image of s in the stalk $\mathcal{F}_q$. This is a functor which is left-adjoint to the forgetful functor from sheaves on X to presheaves on X , inducing an isomorphism of stalks.

Suppose $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves of sets on a topological space X . The following are equivalent statements. If these conditions hold $\mathcal{F}$ is said to be a *subsheaf* of $\mathcal{G}$ where the inclusion ϕ is sometimes left implicit.

- ϕ is a monomorphism in the category of sheaves
- ϕ is injective on the level of stalks ($\phi_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$ is injective for all $p \in X$)
- ϕ is injective on the level of open sets ($\phi(U) : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ is injective for all open $U \in X$)

The following are equivalent; if they hold, we say $\mathcal{G}$ is a *quotient sheaf* of $\mathcal{F}$:

- ϕ is an epimorphism in the category of sheaves
- ϕ is surjective on the level of stalks: $\phi_p : \mathcal{F}_p \rightarrow \mathcal{G}_p$ is surjective for all $p \in X$

Recall that the *base of a topology* is subcollection of open sets $\{B_j\} \subset \{U_i\}$ (with $U_i \subset X$ open), such that each U_i is a union of the B_j . Suppose we have a sheaf $\mathcal{F}$ on a topological space X and a base $\{B_i\}$ of open sets on X . Then consider the information

$$(\{\mathcal{F}(B_i)\}, \{res_{B_i, B_j} : \mathcal{F}(B_i) \rightarrow \mathcal{F}(B_j)\})$$

which is a subset of the information contained in the sheaf—we are only paying attention to the information involving elements of the base, not all open sets. We can recover the entire sheaf because we can determine stalks and then determine when the germs are compatible. We make this explicit. From $x \in X$, define the stalk from the basis data:

$$\mathcal{F}_x := \varinjlim_{x \in B_i \in \{B_i\}} \mathcal{F}(B_i)$$

For an open $U \subseteq X$, set

$$\mathcal{F}(U) = \left\{ (s_x)_{x \in U} \in \prod_{x \in U} \mathcal{F}_x \mid \forall x \in U, \exists B_i \in \{B_i\}, x \in B_i \subseteq U, \exists t \in \mathcal{F}(B_i) \text{ with } t_y = s_y \forall y \in B_i \right\}$$

That is, a section over U is a family of germs, one at each point, locally coming from an actual basis section. For the restrictions, given $V \subseteq U$, just forget the germs outside V .

This suggests the notion of the *sheaf on a base*; a sheaf of sets (or abelian groups, rings, ...) on the base $\{B_i\}$ is the following. For each B_i in the base, we have a set $F(B_i)$. If $B_i \subset B_j$, we have maps $res_{B_j, B_i} : F(B_j) \rightarrow F(B_i)$ with $res_{B_i, B_i} = id_{F(B_i)}$. If $B_i \subset B_j \subset B_k$, then $res_{B_k, B_i} = res_{B_j, B_i} \circ res_{B_k, B_j}$. So far we have defined a presheaf on a base $\{B_i\}$. We require the base identity axiom: If $B = \cup B_i$, then if $f, g \in F(B)$ are such that $res_{B, B_i} f = res_{B, B_i} g$ for all i , then $f = g$. We require the base gluability axiom too: if $B = \cup B_i$, and we have $f_i \in F(B_i)$ such that f_i agrees with f_j on any basic open set contained in $B_i \cap B_j$ (i.e. $res_{B_i, B_k} f_i = res_{B_j, B_k} f_j$) for all $B_k \subset B_i \cap B_j$) then there exists $f \in F(B)$ such that $res_{B, B_i} f = f_i$ for all i .

Suppose $\{B_i\}$ is a base on X , F a sheaf of sets on this base. Then there is a sheaf $\mathcal{F}$ unique up to a unique isomorphism extending F with isomorphisms $\mathcal{F}(B_i) \xrightarrow{\sim} F(B_i)$ agreeing with the restriction maps.

A morphism $F \rightarrow G$ of sheaves on the base is a collection of maps $F(B_k) \rightarrow G(B_k)$ such that the following commutes for all $B_j \hookrightarrow B_i$; a morphism of sheaves is induced by the induced morphism of sheaves on the base, and the morphism of sheaves on the base gives a morphism of the induced sheaves.

$$\begin{array}{ccc} \mathcal{F}(B_i) & \longrightarrow & \mathcal{G}(B_i) \\ \downarrow res_{B_i, B_j} & & \downarrow res_{B_i, B_j} \\ \mathcal{F}(B_j) & \longrightarrow & \mathcal{G}(B_j) \end{array}$$

Suppose $X = \cup U_i$ is an open cover of X , and we have sheaves $\mathcal{F}_i$ on U_i along with isomorphisms $\phi_{ij} : \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j}$ (with ϕ_{ii} the identity) that agree on triple overlaps, i.e. $\phi_{jk} \circ \phi_{ij} = \phi_{ik}$ on $U_i \cap U_j \cap U_k$. This is called the *cocycle condition*. These sheaves can then be glued together into a sheaf $\mathcal{F}$ on X (unique up to unique isomorphism) with isomorphisms $\mathcal{F}|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$ and the isomorphisms over $U_i \cup U_j$ are the obvious ones. We show this explicitly; we can define the presheaf $\mathcal{F}$ on X by

$$\mathcal{F}(U) := \{(s_i)_i | s_i \in \mathcal{F}_i(U \cap U_i), \text{ and } \phi_{ij}(s_i|_{U \cap U_i \cap U_j}) = s_j|_{U \cap U_i \cap U_j} \text{ for all } i, j\}$$

Sections over U are families of local sections that agree on overlaps via the ϕ_{ij} . We can show that this is a sheaf. Moreover, for any $U \subseteq U_i$, define $\alpha_i(U) : \mathcal{F}(U) \rightarrow \mathcal{F}_i(U)$ by $(s_j)_j \mapsto s_i$; this is well-defined and natural in U , and it is an isomorphism with inverse $\beta_i(U) : \mathcal{F}_i(U) \rightarrow \mathcal{F}(U)$, defined by $t \mapsto (t_j)_j$, where $t_i := t$ and $t_j := \phi_{ij}(t|_{U \cap U_j})$. The cocycle condition guarantees the family $(t_j)_j$ satisfies the overlap inequalities, and one checks that their compositions are each the identity, hence $\mathcal{F}|_{U_i} \cong \mathcal{F}_i$, and on $U_i \cap U_j$ the induced identification equals ϕ_{ij} . If $\mathcal{G}$ is another sheaf on X with isomorphisms $\mathcal{G}|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$ compatible on overlaps (i.e., reproducing ϕ_{ij}) then for each $U \subseteq X$ the map $\mathcal{G}(U) \rightarrow \mathcal{F}(U)$ defined by $s \mapsto (\psi_i(s|_{U \cap U_i}))_i$ with $\psi_i : \mathcal{G}|_{U_i} \xrightarrow{\sim} \mathcal{F}_i$ is a natural isomorphism of sheaves; any two such isomorphisms agreeing on the U_i coincide by sheaf locality, so the isomorphism is unique.

The stalk of the kernel is the kernel of the stalks: for all $p \in X$, there is a natural isomorphism

$$(\ker(\mathcal{F} \rightarrow \mathcal{G}))_p \xrightarrow{\sim} \ker(\mathcal{F}_p \rightarrow \mathcal{G}_p)$$

Consider $\phi : \mathcal{F} \rightarrow \mathcal{G}$, namely its cokernel in the category of presheaves, which we call $\mathcal{H}_{pre}$; let $sh : \mathcal{H}_{pre} \rightarrow \mathcal{H}$ be its sheafification. The composition $\mathcal{G} \rightarrow \mathcal{H}$ is the cokernel of ϕ in the category of sheaves. The stalk of the cokernel is naturally isomorphic to the cokernel of the stalk.

The image sheaf $im\phi$ of a morphism of sheaves of abelian groups $\phi : \mathcal{F} \rightarrow \mathcal{G}$ is the sheafification of the image presheaf. As a consequence, the exactness of a sequence of sheaves may be checked

at the level of stalks; i.e., for $\alpha : \mathcal{F} \rightarrow \mathcal{G}$ and $\beta : \mathcal{G} \rightarrow \mathcal{H}$ two morphisms of sheaves of abelian groups on X , $\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$ is exact at $\mathcal{G}$ if and only if $\mathcal{F}_p \xrightarrow{\alpha} \mathcal{G}_p \xrightarrow{\beta} \mathcal{H}_p$ is exact for all $p \in X$. Taking the stalk of a sheaf of abelian groups is an exact functor; more precisely, if X is a topological space, taking the stalk at $p \in X$ defined an exact functor $Ab_X \rightarrow Ab$.

Suppose $U \subset X$ is open, and $0 \rightarrow \mathcal{F} \rightarrow \mathcal{G} \rightarrow \mathcal{H}$ is an exact sequence of sheaves of abelian groups. Then $0 \rightarrow \mathcal{F}(U) \rightarrow \mathcal{G}(U) \rightarrow \mathcal{H}(U)$ is exact. Moreover, if $\pi : X \rightarrow Y$ is a continuous map, then $0 \rightarrow \pi_* \mathcal{F} \rightarrow \pi_* \mathcal{G} \rightarrow \pi_* \mathcal{H}$ is exact.

If $(X, \mathcal{O}_X)$ a ringed space then $\mathcal{O}_X$ modules form an abelian category.

Suppose $\mathcal{O}_X$ is a sheaf of rings on X . The *tensor product* of two $\mathcal{O}_X$ -modules is defined in EXERCISE 2.6.K which I should do. The tensor product of stalks is the stalk of the tensor product.

For some continuous map $\pi : X \rightarrow Y$ and a sheaf $\mathcal{F}$ on X , we have defined the pushforward sheaf $\pi_* \mathcal{F}$ on Y ; the notion of an *inverse image sheaf* comes from a covariant functor π^{-1} from sheaves on Y to sheaves on X , defined to be the left adjoint to π_* . The inverse image of $\mathcal{G}$ is then defined to be $\pi^{-1} \mathcal{G} := (\pi_{pre}^{-1} \mathcal{G})^{sh}$

Suppose (INSERT PUSH PULL MAP FROM 2.7.4.1) is a commutative diagram, and $\mathcal{F}$ a sheaf on X . Define the *push-pull map*

$$\beta^{-1} \alpha_* \mathcal{F} \rightarrow \alpha'_* (\beta')^{-1} \mathcal{F}$$

of sheaves on Y as follows: Start with the identity $(\beta')^{-1} \mathcal{F} \xrightarrow{\sim} (\beta')^{-1} \mathcal{F}$ on W ; by adjointness of $((\beta')^{-1}, \beta'_*)$, this is the same as the data of a morphism $\mathcal{F} \rightarrow (\beta'_*) (\beta')^{-1} \mathcal{F}$ on X . Apply α_* to get a map $\alpha_* \mathcal{F} \rightarrow \alpha_* (\beta'_*) (\beta')^{-1} \mathcal{F}$ on Z . By commutativity, this is the map $\alpha_* \mathcal{F} \rightarrow \beta_* (\alpha'_*) (\beta')^{-1} \mathcal{F}$ on Z . By adjointness of (β^{-1}, β_*) this yields the map.

Suppose $\mathcal{F}$ is a sheaf (or separated presheaf) of abelian groups on X , and s is a global section of $\mathcal{F}$. Define the *support of the section* s , $\text{Supp } s$, to be the set of points p of X where s has a nonzero germ:

$$\text{Supp } s := \{p \in X : s_p \neq 0 \text{ in } \mathcal{F}_p\}$$

. We think of this as the subset of X where the section s lives; the complement is the locus where s is the 0-section. This is a closed subset of X . The *support of a sheaf* of groups $\mathcal{G}$, denoted $\text{Supp } \mathcal{G}$, is the locus where the stalks are nontrivial:

$$\text{Supp } \mathcal{G} := \{p \in X : |\mathcal{G}_p| \neq 1\}$$

Equivalently, $\text{Supp } \mathcal{G}$ is the union of the supports of sections over all open sets. This is a stalk-local notion and hence commutes with restriction to open sets.

2 The spectrum of a ring

A *scheme* consists of the following:

- The set: points of the scheme
- The topology: the open sets of the scheme
- The structure sheaf: the sheaf of "algebraic functions" (sheaf of rings) on the scheme

We require for the topology that the subset where an algebraic function vanishes must be closed.

Definition. The *spectrum* of a ring A , denoted $\text{Spec } A$, is the set of prime ideals of A . Elements $a \in A$ will be called *functions* on $\text{Spec } A$ and their value at the point $[\mathfrak{p}]$ will be $a \pmod{\mathfrak{p}}$. An element of a ring lying in a prime ideal $\mathfrak{p}$ translates to a function that is 0 (i.e. vanishing) at the point $[\mathfrak{p}]$.

If A is generated over a base field (or base ring) by elements $x_1, \dots, x_r$, those elements are often called coordinates because we will later be able to reinterpret them as restrictions of "coordinates on r -space".

Example. The *complex affine line* $\mathbb{A}_{\mathbb{C}}^1 := \text{Spec } \mathbb{C}[x]$. $f(x) = x^2 - 3x + 1$ is a function for example; its value at $[(x-1)]$ which we think of as the point $1 \in \mathbb{C}$ is just $f(1)$. At $[(0)]$ it's $f(x) \pmod{0} = f(x)$.

We can equally define an affine line over any algebraically closed field k .

Example. Consider the set $\text{Spec } k[\epsilon]/(\epsilon^2)$, the spectrum of the ring of dual numbers. We think of ϵ as a very small number so small that its square is zero (although itself is not zero); it is a nonzero function whose value at all points is zero. The elements of the ring of dual numbers thus look like $a + b\epsilon$ for $a, b \in k$. Its prime ideals correspond to the set of prime ideals containing (ϵ^2) . Only prime ideal containing this is (ϵ) so $\text{Spec } k[\epsilon]/(\epsilon^2)$ is just one point (ϵ) .

Example. Consider the set $\text{Spec } k[x]_{(x)}$. Recall notation: for some ring A and some prime ideal $\mathfrak{p}$, $A_{\mathfrak{p}}$ means you're allowed to divide by elements not in $\mathfrak{p}$. The only primes of $k[x]$ in (x) are (0) and (x) , so after localization they are the only ideals that remain prime, therefore $\text{Spec } k[x]_{(x)} = \{(0), (x)k[x]_{(x)}\}$.

Example. For the affine line over $\mathbb{R}$, since $\mathbb{R}[x]$ is a Euclidean domain we have prime ideals (0) , $(x-a)$ where $a \in \mathbb{R}$, and (x^2+ax+b) where x^2+ax+b is an irreducible quadratic. The latter two are maximal ideals i.e. their quotients are fields. For example $\mathbb{R}[x]/(x-3) \cong \mathbb{R}$, $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$.

Example. Consider the affine line over the rationals $\mathbb{Q}$, i.e. $\text{Spec } \mathbb{Q}[x]$. Because $\mathbb{Q}$ is a field, $\mathbb{Q}[x]$ is a PID. (0) is of course a prime ideal; nonzero prime ideals are irreducible polynomials in $\mathbb{Q}[x]$.

In general, we define *affine n -space* $\mathbb{A}_A^n$ to be $\text{Spec } A[x_1, \dots, x_n]$, where A is an arbitrary ring.

Theorem. (*Weak Nullstellensatz*) If k is algebraically closed field, the maximal ideals of $k[x_1, \dots, x_n]$ are precisely those ideals of the form $(x_1 - a_1, \dots, x_n - a_n)$ where $a_i \in k$.

Theorem. (*Nullstellensatz*) If k is any field, every maximal ideal of $k[x_1, \dots, x_n]$ has residue field a finite extension of k (i.e. any field extension of k that is finitely generated as a k -algebra is necessarily also finitely generated as a k -vector space, i.e. is a finite extension of fields.)

Prop. Let A be a ring, $I \subseteq A$ an ideal. There is an inclusion-preserving bijection between $\text{Spec } A/I$ and $\{\mathfrak{p} \in \text{Spec } A : I \subseteq \mathfrak{p}\}$ given by $\mathfrak{q} \mapsto \phi^{-1}(\mathfrak{q})$, $\mathfrak{p} \mapsto \mathfrak{p}/I$, where $\phi : A \rightarrow A/I$ is the quotient map.

Prop. Suppose S is a multiplicative subset of A . There is an order-preserving bijection of the prime ideals of $S^{-1}A$ with the prime ideals of A that don't meet the multiplicative set S .

Prop. Ring homomorphisms induce maps of spectra (as sets); a map of rings $\phi : B \rightarrow A$ induces a map of sets $\text{Spec } A \rightarrow \text{Spec } B$ "in the opposite direction", giving a contravariant functor from the category of rings to the category of sets: the composition of two maps of rings induces the composition of the corresponding maps of spectra.

Functions are not determined by their values at points. Ring elements that have a power that is 0 are called nilpotents. If I is an ideal of nilpotents then $\text{Spec } B/I \hookrightarrow \text{Spec } B$ is a bijection. The nilpotents form an ideal called the *nilradical* $\mathfrak{N}(B)$, which is the intersection of all prime ideals of B ; geometrically, a function on $\text{Spec } A$ vanishes at every point if and only if it is nilpotent. Two functions have the same values at points if and only if they differ by a nilpotent; geometrically, a function vanishes at every point of the spectrum of a ring if and only if it has a power that is zero, and if there are no nonzero nilpotents, then functions are determined by their values at points. Such a ring with zero nilpotents is said to be *reduced*.

Definition. If S is a subset of a ring A define the *vanishing set* of A by $V(S) := \{[p] \in \text{Spec } A : S \subset p\}$. We define the *Zariski topology* by declaring that $V(S)$ is closed for all S .

$\text{Spec } B/I$ is a closed subset of $\text{Spec } B$. If $S = \{1, f, f^2, \dots\}$, $\text{Spec } S^{-1}B$ is an open subset of $\text{Spec } B$, but for arbitrary S the spectrum of the localization need not be open or closed. The Zariski topology on $\text{Spec } B/I$ is the subspace topology induced by inclusion in $\text{Spec } B$.

If $I \subset A$ is an ideal define its *radical* $\sqrt{I} = \{r \in A : r^n \in I \text{ for some } n \in \mathbb{N}\}$. An ideal is said to be radical if $\sqrt{I} = I$. Prime ideals are radical. If $I_1, \dots, I_n$ are ideals of A , then $\sqrt{\bigcap_{i=1}^n I_i} = \bigcap_{i=1}^n \sqrt{I_i}$. $\sqrt{I}$ is the intersection of all the prime ideals containing I .

Prop. Suppose $I \subset B$ is an ideal. Then f vanishes on $V(I)$ if and only if $f \in \sqrt{I}$.

Definition. If $f \in A$ define the *distinguished open set* $D(f) := \{[p] \in \text{Spec } A : f \notin p\} = \{[p] \in \text{Spec } A : f([p]) \neq 0\}$. It is the locus where f doesn't vanish. These form a base for the Zariski topology.

Prop. If $f_i \in A$ as i runs over some index J , then $\bigcup_{i \in J} D(f_i) = \text{Spec } A$ if and only if $(\{f_i\}_{i \in J}) = A$, or equivalently and very usefully, if there are a_i ($i \in J$), all but finitely many 0, such that $\sum_{i \in J} a_i f_i = 1$.

Prop. If $\text{Spec } A$ is an infinite union of distinguished open sets $\bigcup_{j \in J} D(f_j)$ then it is a union of a finite number of these, i.e. there is a finite subset J' such that $\text{Spec } A = \bigcup_{j \in J'} D(f_j)$.

Prop. $D(f) \cap D(g) = D(fg)$. $D(f) \subset D(g)$ if and only if $f^n \in (g)$ for some $n \geq 1$, if and only if g is an invertible element of A_f .

- Recall that a topological space X is connected if it cannot be written as the disjoint union of two nonempty open sets. $\text{Spec } A$ is not connected if and only if A is isomorphic to the product of nonzero rings A_1 and A_2 .
- A topological space is said to be irreducible if it is nonempty and it is not the union of two proper closed subsets. In such a topological space, any nonempty open set is dense, and subsets of the topological space with subspace topology are irreducible if and only if their closure in the topological space is irreducible. An irreducible topological space is connected.
- A topological space X is *quasicompact* if given any cover by open sets $\bigcup_{i \in I} U_i$, there is a finite subset $S \subseteq I$ such that $X = \bigcup_{i \in S} U_i$. $\text{Spec } A$ is quasicompact, but can have nonquasicompact open sets.
- A point of a topological space $p \in X$ is said to be a *closed point* if $\{p\}$ is a closed subset. The closed points of $\text{Spec } \bar{k}[x_1, \dots, x_n]/(f_1, \dots, f_r)$ (where $\bar{k}$ is alg closed field) are naturally interpreted as those points in $\bar{k}^n$ satisfying the equations $f_1 = \dots = f_r = 0$.

- Given two points x, y of a topological space X , we say that x is a *specialization* of y and y is a *generalization* of x if $x \in \overline{\{y\}}$. If $X = \operatorname{Spec} A$ then $[q]$ is a specialization of $[p]$ if and only if $p \subset q$, hence $V(p) = \overline{\{[p]\}}$.
- We say that a point $p \in X$ is a *generic point* for a closed subset K if $\overline{\{p\}} = K$. If something is "generically" or "mostly" true for the points of an irreducible subset in the sense of being true for a dense open subset (for "almost all points") then it is true for the generic point and vice versa. The phrase general point does not have the same meaning; "the general point of X satisfies such-and-such a property" means "every point of some dense open subset of X satisfies such-and-such a property".
- An irreducible component of a topological space is a maximal irreducible subset (an irreducible subset not contained in any larger irreducible subset). Irreducible components are closed and it can be helpful to think of them as maximal among the irreducible closed subsets of X . Similarly a subset Y of a topological space X is a connected component if it is a maximal connected subset (a connected subset not contained in any larger connected subset). Every topological space is the union of irreducible components. Every point is contained in a connected component and connected components are always closed.

Given a subset $S \subset \operatorname{Spec} A$, $I(S)$ is the set of functions vanishing on S , i.e. $I(S) = \bigcap_{[p] \in S} \mathfrak{p} \subset A$ when S is nonempty.

Prop.

- $I(S)$ is an ideal of A
- $I(\cdot)$ is inclusion reversing: $S_1 \subset S_2$ implies $I(S_2) \subset I(S_1)$
- $I(\overline{S}) = I(S)$
- $V(I(S)) = \overline{S}$, hence $V(I(S)) = S$ for closed S

Prop. $V(\cdot)$ and $I(\cdot)$ give an inclusion-reversing bijection between closed subsets of $\operatorname{Spec} A$ and radical ideals of A (where a closed subset gives a radical ideal by $I(\cdot)$, and a radical ideal gives a closed subset by $V(\cdot)$). They moreover give a bijection between irreducible closed subsets of $\operatorname{Spec} A$ and prime ideals of A , so in $\operatorname{Spec} A$ there is a bijection between points of $\operatorname{Spec} A$ and irreducible closed subsets of $\operatorname{Spec} A$ where a point determines an irreducible closed subset by taking the closure. Hence each irreducible closed subset of $\operatorname{Spec} A$ has precisely one generic point, any irreducible closed subset can be written uniquely as $\overline{\{z\}}$.

Prop. The irreducible components of $\operatorname{Spec} A$ are in bijection with the minimal prime ideals of A ; in particular $\operatorname{Spec} A$ is irreducible iff A only has one minimal prime ideal.

3 Schemes

Definition. Define $\mathcal{O}_{\operatorname{Spec} A}(D(f))$ to be the localization of A at the multiplicative set of all functions that do not vanish outside of $V(f)$ (i.e. those $g \in A$ such that $V(g) \subset V(f)$ or equivalently $D(f) \subset D(g)$).

Prop. The natural map $A_f \rightarrow \mathcal{O}_{\operatorname{Spec} A}(D(f))$ is an isomorphism.

If $D(f') \subset D(f)$ define the restriction map $\text{res}_{D(f), D(f')} : \mathcal{O}_{\text{Spec } A}(D(f)) \rightarrow \mathcal{O}_{\text{Spec } A}(D(f'))$ in the obvious way: the latter ring is a further localization of the former ring. The data just described give a sheaf on the distinguished base and hence determine a sheaf on the topological space $\text{Spec } A$, which we call the *structure sheaf*.

Definition. An *affine scheme* is a topological space with a structure sheaf. More precisely, an affine scheme is a ringed space isomorphic to $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ for some A . A *scheme* $(X, \mathcal{O}_X)$ is a ringed space such that any point of X has an open neighborhood U such that $(U, \mathcal{O}_X|_U)$ is an affine scheme. The topology on a scheme is called the *Zariski topology*.

An isomorphism of schemes is an isomorphism as ringed spaces. If $U \subset X$ is an open subset, then the elements of $\Gamma(U, \mathcal{O}_X)$ are said to be the *functions* on U , generalizing the definitions of functions on an affine scheme.

Prop. If $(X, \mathcal{O}_X)$ is an affine scheme, we may recover its ring by taking the ring of global sections as $X = D(1)$.

If X is a scheme and U is any open subset, $(U, \mathcal{O}_X|_U)$ is also a scheme. We call this the *open subscheme* of X ; if U is also an affine scheme, we often say that U is an *affine open subset*, or an *affine open subscheme*. For example, $D(f)$ is an affine open subscheme of $\text{Spec } A$. The affine open sets form a base for the Zariski topology.

We say that a ringed space is a *locally ringed space* if its stalks are local rings. The stalk of $\mathcal{O}_{\text{Spec } A}$ at the point $[p]$ is the local ring A_p .

Schemes, manifolds are examples of locally ringed spaces; taking the quotient by a maximal ideal may be interpreted as evaluating at the point.

The maximal ideal of the local ring $\mathcal{O}_{x,p}$ is denoted $m_{X,p}$ or m_p and the residue field $\mathcal{O}_{X,p}/m_p$ is denoted $\kappa(p)$. Each function f on an open subset of U of a locally ringed space X has a value at each point p of U : the value of f at p , denoted $f(p)$, is the image of the germ of f at P under the canonical map $\mathcal{O}_{X,p} \rightarrow \kappa(p)$. The residue field at $[p]$ is A_p/pA_p which is isomorphic to the fraction field of the quotient $K(A/p)$.

Prop. The following commutes.

$$\begin{array}{ccc}
 & A_p & \\
 \text{localize} \nearrow & & \searrow \text{quotient} \\
 A & & A_p/pA_p = K(A/p) \\
 \searrow \text{quotient} & & \nearrow \text{localize} \\
 & A/p &
 \end{array}$$

If $\mathcal{F}$ is an $\mathcal{O}_x$ -module on a scheme X (or more generally a locally ringed space) define the *fiber* of $\mathcal{F}$ at a point $p \in X$ by $\mathcal{F}|_p := \mathcal{F}_p \otimes_{\mathcal{O}_{x,p}} \kappa(p)$. The fiber is a vector space over $\kappa(p)$. The dimension of this vector space is called the *rank* of $\mathcal{F}$ at p . For example, $\mathcal{O}_X|_p$ is $\kappa(p)$ and $\text{rank}_p \mathcal{O}_X = 1$.

Given two topological spaces X and Y , and open subsets $U \subset X$ and $V \subset Y$ along with a homeomorphism $U \xrightarrow{\sim} V$, we can create a new topological space W , that we think of as gluing X and Y along $U \xrightarrow{\sim} V$. It is the quotient of the disjoint union $X \sqcup Y$ by the equivalence relation $U \sim V$ where the quotient is given by the quotient topology. Then X and Y are naturally identified with open subsets of W and indeed cover W . Similarly, for two schemes $(X, \mathcal{O}_X)$, $(Y, \mathcal{O}_Y)$ and open

subsets $U \subset X$ and $V \subset Y$ along with a homeomorphism $\pi : U \xrightarrow{\sim} V$ and an isomorphism of structure sheaves $\mathcal{O}_V \xrightarrow{\sim} \pi_* \mathcal{O}_U$ (i.e. an isomorphism of schemes $(U, \mathcal{O}_U|_U) \xrightarrow{\sim} (V, \mathcal{O}_V|_V)$) then we can glue these together to get a single schemes.

Prop. Given schemes X_i where i runs over some index set I , and open subschemes $X_{ij} \subset X_i$, with $X_{ii} = X_i$, and isomorphisms $f_{ij} : X_{ij} \xrightarrow{\sim} X_{ji}$ with f_{ii} the identity, such that the cocycle condition holds (i.e. $f_{ik}|_{X_{ij} \cap X_{ik}} = f_{jk}|_{X_{ji} \cap X_{jk}} \circ f_{ij}|_{X_{ij} \cap X_{ik}}$) then we can show that there is a unique scheme X (up to a unique isomorphism) along with open subschemes isomorphic to the X_i respecting this gluing data.

Example. *Projective n -space over a field k* is denoted $\mathbb{P}_k^n$; we construct it by gluing together $n+1$ open sets each isomorphic to $\mathbb{A}_k^n$. For $\mathbb{P}_k^n$, we glue together $n+1$ open sets, one for each of $i = 0, \dots, n$. The i th open set U_i will have coordinates $x_{0/i}, \dots, x_{(i-1)/i}, x_{(i+1)/i}, \dots, x_{n/i}$. It will be convenient to write as $\text{Spec } k[x_{0/i}, x_{1/i}, \dots, x_{n/i}]/(x_{i/i} - 1)$ (we have introduced a dummy variable $x_{i/i}$ which we immediately set to 1). We glue the distinguished open set $D(x_{j/i})$ of U_i to the distinguished open set $D(x_{i/j})$ of U_j , by identifying these two schemes by describing the identification of rings $k[x_{0/i}, x_{1/i}, \dots, x_{n/i}, 1/x_{j/i}]/(x_{i/i} - 1) \cong k[x_{0/j}, x_{1/j}, \dots, x_{n/j}, 1/x_{i/j}]/(x_{j/j} - 1)$ via $x_{k/i} = x_{k/j}/x_{i/j}$ and $x_{k/j}/x_{j/i}$ (which implies $x_{i/j}x_{j/i} = 1$). Notice that this does not require k is a field, so we might as well define $\mathbb{P}_A^n$ for any ring A .

Example. $\mathbb{A}_k^n$ was shown to be irreducible; this argument behaves well under gluing, yielding that $\mathbb{P}_k^n$ is irreducible.

Prop. There is a bijection between irreducible closed subsets and points for schemes in general.

If X is a scheme that has a finite cover $X = \cup_{i=1}^n \text{Spec } A_i$ where A_i is Noetherian, then X is a Noetherian topological space. $\mathbb{P}_k^n, \mathbb{P}_{\mathbb{Z}}^n$ are Noetherian topological spaces built by gluing a finite number of spectra of Noetherian rings.

Definition. Scheme X is *quasicompact* iff it can be written as a finite union of affine open subschemes (hence $\mathbb{P}_A^n$ is quasicompact for any ring A).

Every point of X has a closed point in its closure. Every nonempty closed subset of X contains a closed point X . In particular, every nonempty quasicompact scheme has a closed point.

If there is some property P of points of a scheme that is "open" (if point p has P then there is some open neighborhood of p such that all point in it have P), then to check if all of a quasicompact scheme have P it suffices to check only the closed points. The points of traditional varieties are the closed points of the corresponding schemes.

Definition. A topological space is *quasiseparated* if the intersection of any two quasicompact open subsets is quasicompact. A scheme is *quasiseparated* iff the intersection of any two affine open subsets is a finite union of affine open subsets, iff there exists a cover of it by affine open subsets, any two of which have intersection also covered by a finite number of affine open subsets. Affine schemes are quasiseparated.

A scheme is quasicompact and quasiseparated iff it can be covered by a finite number of affine open subsets, any two of which have intersection also covered by a finite number of affine open subsets. People call this *qcqs* for short.

Prop. All projective A -schemes are qcqs.

Definition. A scheme X is *reduced* if its stalks $\mathcal{O}_{X,p}$ have no nonzero nilpotents (for all p).

If f and g are two functions (global sections of $\mathcal{O}_X$) on a reduced scheme that agree at all points then $f = g$.

A ring is said to be reduced if it has no nonzero nilpotents. A is a reduced ring iff $\text{Spec } A$ is reduced. A scheme X is reduced if $\mathcal{O}_X(U)$ is reduced for every open set U of X .

Prop. $\mathbb{A}_k^n, \mathbb{P}_k^n$ are reduced.

Reducedness is not in general an open condition.

Prop. If X is quasicompact and f is a function that vanishes at all points of X , then there is some n such that $f^n = 0$, and this may fail if X is not quasicompact.

Definition. A scheme X is *integral* if it is nonempty, and $\mathcal{O}_X(U)$ is an integral domain for every nonempty open subset U of X .

A scheme X is integral iff it is irreducible and reduced. An affine scheme $\text{Spec } A$ is integral iff A is an integral domain.

An integral scheme X , being irreducible, has a generic point η ; suppose $\text{Spec } A$ is any nonempty affine open subset of X , then $\mathcal{O}_{X,\eta}$ (the stalk of $\mathcal{O}_X$ at η) is naturally identified with the fraction field of A , $K(A)$; this is called the *function field* $K(X)$ of X , which can be computed on any nonempty open set, as any such open set contains the generic point. The elements of $K(X)$ are called *rational functions* on X .

Restriction maps on an integral scheme are inclusions so long as the open subset that you're restricting to is nonempty. For any nonempty affine open subset of X , $\text{Spec } A$ (so A is integral domain), the natural map $\mathcal{O}_X(U) \rightarrow \mathcal{O}_{X,\eta} = K(A)$ is an inclusion. So for irreducible varieties, sections over different open sets can be considered subsets of the same ring.

Integrality is not stalk-local (the disjoint union of two integral schemes is not integral)

Prop. Suppose $\text{Spec } A, \text{Spec } B$ are affine open subschemes of a scheme X ; then the union of these subschemes is the union of open sets that are simultaneously distinguished open subschemes of $\text{Spec } A$ and $\text{Spec } B$.

Lemma. (*Affine communication*) Let P be some property enjoyed by some affine open subsets of a scheme X , such that for any affine open subset $\text{Spec } A \hookrightarrow X$,

- If $\text{Spec } A \hookrightarrow X$ has property P then for any $f \in A$, $\text{Spec } A_f \hookrightarrow X$ does, too.
- If $(f_1, \dots, f_n) = A$ (i.e. the $\text{Spec } A_{f_i}$ cover $\text{Spec } A$) and $\text{Spec } A_{f_i} \hookrightarrow X$ has P for all i , then so does $\text{Spec } A \hookrightarrow X$

Suppose that $X = \cup_{i \in I} \text{Spec } A_i$ where $\text{Spec } A_i$ has property P . Then every affine open subset of X has P too.

We say that such a property is *affine-local* (i.e. a property is affine-local if we can check it on any affine cover).

Note that if U is an open subscheme of X then it inherits any affine-local property of X .

Any property that is stalk-local (a scheme has property P iff all its stalks have property Q) is necessarily affine-local (a scheme has property P iff all of its affine open sets have property R , where an affine scheme has property R iff all its stalks have property Q).

Suppose A is a ring, and $(f_1, \dots, f_n) = A$.

- If A is Noetherian, then so is A_{f_i} ; if each A_{f_i} is Noetherian, then so is A .
- Suppose B is a ring and A is a B -algebra, hence A_g is a B -algebra for all $g \in A$. If A is a finitely generated B -algebra, then so is A_{f_i} ; if each A_{f_i} is a finitely generated B -algebra, then so is A .

Definition. A scheme X is said to be *locally Noetherian* if it can be covered by affine open sets $\text{Spec } A$ where A is Noetherian. X is said to be *Noetherian* if it is in addition quasicompact or equivalently can be covered by finitely many such affine open sets.

Locally Noetherian schemes are quasiseparated. Noetherian schemes have a finite number of irreducible components and a finite number of connected components, each a finite union of irreducible components.

Prop. A Noetherian scheme X is integral iff it is nonempty and connected and all stalks $\mathcal{O}_{X,p}$ are integral domains.

Definition. An A -scheme, or a *scheme over A* , where A is a ring, is a scheme where all the rings of sections of the structure sheaf (over all open sets) are A -algebra, and all restriction maps are maps of A -algebras.

Suppose X is an A -scheme. If X can be covered by affine open sets $\text{Spec } B_i$ where each B_i is a finitely generated A -algebra, X is said *locally of finite type over A* , or a *locally finite type A -scheme*. If it is furthermore quasicompact it is of *finite type over A* or a *finite type A -scheme*. A scheme locally of finite type over k or $\mathbb{Z}$ or any Noetherian ring is locally Noetherian, and a scheme of finite type over any Noetherian ring is Noetherian.

A quasiprojective A -scheme is of finite type over A ; if A is furthermore assumed to be Noetherian, the A -scheme is Noetherian, and hence has a finite number of irreducible components. An open subscheme of a projective A -scheme is locally of finite type over A , and if A is Noetherian, then it is quasicompact and hence quasiprojective and hence of finite type over A .

Definition. An affine scheme that is reduced and of finite type over k is said to be an *affine variety* over k or an *affine k -variety*; a reduced (quasi)projective k -scheme is a *(quasi)projective variety* (over k), or a *(quasi)projective k -variety*.

A point of a locally finite type k -scheme is a closed point iff the residue field of the stalk of the structure sheaf at that point is a finite extension of k . These closed points are dense.

The *degree of a closed point p* of a locally finite type k -scheme (e.g. a variety over k) is the degree of the field extension $\kappa(p)/k$. For example, in $\mathbb{A}_k^1 = \text{Spec } k[t]$ the point $[(p(t))]$ ($p(t) \in k[t]$ irreducible) is $\deg p(t)$. If k is algebraically closed, the degree of every closed point is 1.

Recall that an integral domain A is integrally closed if the only zeros in $K(A)$ to any monic polynomial in $A[x]$ must lie in A itself.

Definition. A scheme X is *normal* if all of its stalks $\mathcal{O}_{X,p}$ are normal, i.e. are integral domains, and integrally closed in their fraction fields (so by definition normality is a stalk-local property).

As reducedness is a stalk-local property, normal schemes are reduced. A ring is normal if its spectrum is normal as a scheme.

Integrally closed domains behave well under localization: if A is an integrally closed domain and S is a multiplicative subset not containing 0, then $S^{-1}A$ is an integrally closed domain.

A Noetherian scheme is normal iff it is the finite disjoint union of integral Noetherian schemes.

Prop. If A is an integral domain, then the following are equivalent:

- A is integrally closed
- A_p is integrally closed for all prime ideals $p \subset A$
- A_m is integrally closed for all maximal ideals $m \subset A$

If all the stalks of a scheme X are UFDs, we say X is *factorial*. Any nonzero localization of a UFD is a UFD. Thus if A is a UFD then its spectrum is factorial. Factoriality implies normality; UFDs are integrally closed, hence factorial schemes are normal, and if A is a UFD, then $\text{Spec } A$ is normal.

To check if a Noetherian integral domain is a UFD:

- rings with Euclidean algorithm such as $\mathbb{Z}$, $k[t]$, $\mathbb{Z}[i]$; polynomial rings over a UFD, by Gauss' lemma;
- the localization of a UFD is also a UFD;
- height 1 prime ideals are principal;
- normal and $C1 = 0$
- Nagata's lemma

Prop. $\mathbb{A}_k^n, \mathbb{P}_k^n, \text{Spec } \mathbb{Z}$ are all normal.

Example. Suppose A is a UFD with 2 invertible, and $z^2 - f$ is irreducible in $A[z]$.

- If $f \in A$ has no repeated prime factors then $\text{Spec } A[z]/(z^2 - f)$ is normal.
- If $f \in A$ has repeated prime factors, then $\text{Spec } A[z]/(z^2 - f)$ is not normal.

Suppose M is an A -module. Define $\widetilde{M}(D(f))$ to be the localization of M at the multiplicative set of all functions that do not vanish outside of $V(f)$, and define the restriction maps in the analogous way to $\mathcal{O}_{\text{Spec } A}$; this defines a sheaf on the distinguished base, and hence a sheaf on $\text{Spec } A$; then this is an $\mathcal{O}_{\text{Spec } A}$ -module.

Note. In showing the latter we must show that if $(f_i)_{i \in I} = A$,

$$0 \longrightarrow M \longrightarrow \prod_{i \in I} M_{f_i} \longrightarrow \prod_{i \neq j \in I} M_{f_i f_j}$$

is exact; in particular M can be identified with a specific submodule of $\prod_{i \in I} M_{f_i}$. Even though $M \rightarrow M_{f_i}$ may not be an inclusion for any f_i , $M \rightarrow \prod_{i \in I} M_{f_i}$ is an inclusion.

Define the *support of an element* $m \in M$ by $\text{Supp } m := \{[p] \in \text{Spec } A : m_p \neq 0\} \subset \text{Spec } A$, which is the support of m considered as a section of $\widetilde{M}$. $\text{Supp } m$ is a closed subset of $\text{Spec } A$. Similarly, define the *support of an A -module* M by $\text{Supp } M := \text{Supp } \widetilde{M} = \{[p] \in \text{Spec } A : M_p \neq 0\} \subset \text{Spec } A$. $\text{Supp } m$ is closed in $\text{Spec } A$; in fact, $\text{Supp } m = V(\text{Ann}(m))$ where $\text{Ann}(m) = \{a \in A : am = 0\}$.

If X is a scheme, then an $\mathcal{O}_X$ -module $\mathcal{F}$ is a *quasicoherent sheaf* if for every affine open subset $\text{Spec } A \subset X$, $\mathcal{F}|_{\text{Spec } A} \cong \widetilde{M}$ for some A -module M . We define the category of quasicoherent

sheaves on X , $QCoh_X$ by taking the morphisms to be morphisms as $\mathcal{O}_X$ -modules. This is an abelian category.

Suppose P is the property of affine open subschemes $\text{Spec } A \subset X$ that $\mathcal{F}|_{\text{Spec } A} \cong \widetilde{M}$ for some A -module M . then P satisfies the two hypotheses of the affine communication lemma. So the quasicoherence of a $\mathcal{O}$ -module can be checked by checking a collection of affine open sets covering X . There is an equivalence $Mod_A \xrightarrow{\sim} QCoh_{\text{Spec } A}$.

Not every $\mathcal{O}_X$ -module is a quasicoherent sheaf.

An A -module M is said to be *torsion free* if $am = 0$ implies that either a is a zerodivisor in A or $m = 0$; if A is an integral domain, the definition of torsion freeness can be restated as $am = 0$ iff $a = 0$ or $m = 0$. In this case, the torsion submodule M_{tors} consists of those elements of M annihilated by some nonzero element of A . M is torsion-free iff $M_{tors} = 0$. We say a module M over an integral domain A is *torsion* if $M = M_{tors}$, which is equivalent to $M \otimes_A K(A) = 0$. If X is a scheme then an $\mathcal{O}_X$ -module $\mathcal{F}$ is said to be torsion-free if $\mathcal{F}_p$ is a torsion-free $\mathcal{O}_{X,p}$ -module for all p .

A quasicoherent sheaf on a reduced scheme is *torsion* if its stalk at every generic point of every irreducible component is 0.

The *distinguished affine base of a scheme* is the data of the affine open sets and distinguished inclusions. I.e. we remember only some of the open sets (the affine open sets) and only some of the morphisms between them (distinguished morphisms). / Given a sheaf $\mathcal{F}$ on X we get a sheaf on the distinguished affine base. The stalk $\mathcal{F}_p$ is the colimit $\text{colim}(f \in \mathcal{F}(U))$ where the colimit is over all open sets contained in X .

- A sheaf on the distinguished affine base $\mathcal{F}^b$ determines a unique (up to a unique isomorphism) sheaf $\mathcal{F}$ which when restricted to the affine base is $\mathcal{F}^b$
- A morphism of sheaves on a distinguished affine base uniquely determines a morphism of sheaves
- An $\mathcal{O}_X$ -module on the distinguished affine base yields an $\mathcal{O}_X$ -module.

Suppose $\mathcal{F}$ is an $\mathcal{O}_X$ -module and $\text{Spec } A_f \hookrightarrow \text{Spec } A \subset X$ is a distinguished open subscheme of an affine open subscheme of X . Let $\phi : \Gamma(\text{Spec } A, \mathcal{F}) \rightarrow \Gamma(\text{Spec } A_f, \mathcal{F})$ be the restriction map. The source of ϕ is an A -module and the target is an A_f module so by the universal property of localization, ϕ naturally factors as $\Gamma(\text{Spec } A, \mathcal{F}) \xrightarrow{\otimes_A A_f} \Gamma(\text{Spec } A, \mathcal{F})_f \xrightarrow{\alpha} \Gamma(\text{Spec } A_f, \mathcal{F})$.

An $\mathcal{O}_X$ -module $\mathcal{F}$ is quasicoherent iff for each such distinguished $\text{Spec } A_f \hookrightarrow \text{Spec } A$, α is an isomorphism. Equivalently, $\mathcal{F}$ is quasi-coherent iff $\alpha : \Gamma(U, \mathcal{F}) \otimes_A A_f \rightarrow \Gamma(\text{Spec } A_f, \mathcal{F})$ is an isomorphism.

If A is a ring and $f \in A$ then $\mathfrak{R}(A_f) \cong \mathfrak{R}(A)_f$. We can thus construct the *quasicoherent sheaf of nilpotents on any scheme* X ; for any affine open U , we can define $\mathfrak{R}_X(U) = \mathfrak{R}(A)$ and hence $\mathfrak{R}_X|_U \cong \widetilde{\mathfrak{R}(A)}$.

The tensor product in the category of quasicoherent sheaves on X can also be described in terms of affine open sets. Given a section s of $\mathcal{F}$ and a section t of $\mathcal{G}$, we have a section $s \otimes t$ of $\mathcal{F} \otimes \mathcal{G}$.

If X is a scheme and $f \in \Gamma(X, \mathcal{O}_X)$ is a function, let $X_f \subset X$ be a subset on which f doesn't vanish. X_f is open.

Lemma. (*qcqs lemma*) Suppose X is a qcqs scheme, $\mathcal{F}$ a quasicoherent sheaf on X , and $f \in \Gamma(X, \mathcal{O}_X)$ a function on X . Then the restriction map $\text{res}_{X_f \subset X} : \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X_f, \mathcal{F})$ is precisely localization.

There are three natural finite conditions on an A -module; if A is Noetherian they are all the same.

- *Finitely generated*, i.e. $\exists$ surjection $A^{\oplus p} \rightarrow M \rightarrow 0$
- *Finitely presented*, i.e. there exists a finite presentation i.e. an exact sequence $A^{\oplus q} \rightarrow A^{\oplus p} \rightarrow M \rightarrow 0$
- *Coherent*, i.e. finitely generated and for any $A^{\oplus p} \rightarrow M$ (not necessarily surjective!) the kernel is finitely generated.

The coherent A -modules form an abelian subcategory of the category of A -modules.

Finitely generated modules over a principal ideal domain are finite direct sums of cyclic modules.

- If $f \in A$, and M is a finitely generated A -module, show that M_f is a finitely generated A_f -module.
- If $(f_1, \dots, f_n) = A$, and M_{f_i} is a finitely generated A_{f_i} -module for all i , then M is a finitely generated A -module.

Suppose M is a finitely presented A -module (for some surjection from a finite free module, the kernel is finitely generated), and $\phi : A^{\oplus p} \rightarrow M$ is any surjection. Then $\ker \phi$ is finitely generated.

Finite presentation satisfies the hypotheses of the affine communication lemma:

- If $f \in A$, and if M finitely presented A -module, M_f is finitely presented A_f -module.
- If $(f_1, \dots, f_n) = A$, and M_{f_i} finitely presented A_{f_i} -module for all i , M is finitely presented A -module.

Coherence satisfies the hypotheses of affine communication:

- If $f \in A$ and M is a coherent A -module, M_f is a coherent A_f -module.
- If $(f_1, \dots, f_n) = A$, and M_{f_i} is a coherent A_{f_i} -module for all i , then M is a coherent A -module.

A quasicoherent sheaf $\mathcal{F}$ is *finite type* (resp. *finitely presented*, *coherent*) if for every affine open $\text{Spec } A$, $\Gamma(\text{Spec } A, \mathcal{F})$ is a finitely generated (resp. finitely presented, coherent) A -module. Coherent sheaves are always on finite type; on a locally Noetherian scheme, all three notions are the same. Coherent sheaves on X form an abelian category Coh_X .

An object $M \in \mathcal{C}$ is *simple* (or irreducible) if its only subobjects are 0 and itself. A *composition series* for M is a finite filtration $0 = X_0 \subset X_1 \subset \dots \subset X_{n-1} \subset X_n = M$ (proper subset) such that the quotients X_{i+1}/X_i are all simple. If M has a composition series, it is said to have finite length.

Theorem. (*Jordan-Holder*) If M has a finite composition series, then all composition series for M have the same length, and the quotients are all the same, possibly rearranged.

We call the length of any of the composition series for M the *length* of M , denoted $\ell(M)$. If A has finite length as a module over itself, A is an *Artinian ring*. If A is furthermore a local ring, we say that A is an Artin or *Artinian local ring*.

The *length* of a scheme X is the length of the structure sheaf $\mathcal{O}_X$ (as an object of QCoh_X ; recall that the structure sheaf of a scheme is always quasicoherent). A scheme X is *finite length* or *Artinian* if $\mathcal{O}_X$ is finite length.

ADD ASSOCIATED POINTS

4 Morphisms of schemes

Morphisms of schemes $\text{Spec } A \rightarrow \text{Spec } B$ should be the ring maps $B \rightarrow A$, much like how ring maps induce maps of topological spaces in the opposite direction. We motivate this. Let $\pi : X \rightarrow Y$ be map of differentiable manifolds, then a smooth function to Y pulls back to a smooth function on X (i.e. there is a natural map $\Gamma(U, \mathcal{O}_Y) \rightarrow \Gamma(\pi^{-1}(U), \mathcal{O}_X)$ for open subsets $U \subseteq Y$). Behaves well w.r.t. restriction so we have a map of sheaves $\mathcal{O}_Y \rightarrow \pi_* \mathcal{O}_X$.

Definition. A *morphism of ringed spaces* $\pi : X \rightarrow Y$ is a continuous map of topological spaces along with a map $\mathcal{O}_Y \rightarrow \pi_* \mathcal{O}_X$. By adjointness this is the same as the map $\pi^{-1} \mathcal{O}_Y \rightarrow \mathcal{O}_X$.

If $U \subset Y$ is an open subset then there is a natural morphism of ringed spaces $(U, \mathcal{O}_U|_U) \rightarrow (Y, \mathcal{O}_Y)$. If $U \rightarrow Y$ is an isomorphism of U with an open subset V of Y , we are given an isomorphism $(U, \mathcal{O}_U) \xrightarrow{\sim} (V, \mathcal{O}_Y|_V)$ via the isomorphism $U \xrightarrow{\sim} V$ then the resulting map of ringed spaces is called an *open embedding (open immersion) of ringed spaces* and the morphism $U \rightarrow Y$ is written $U \hookrightarrow Y$.

Prop. Suppose $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ are ringed spaces, $X = \cup_i U_i$ is an open cover of X , and we have morphisms of ringed spaces $\pi_i : U_i \rightarrow Y$ that agree on overlaps. There is a unique morphism of ringed spaces $\pi : X \rightarrow Y$ that restricts to π_i on every U_i .

Prop. Given a morphism of ringed spaces $\pi : X \rightarrow Y$, the sheaf pushforward induces a functor $\pi_* : \text{Mod}_{\mathcal{O}_X} \rightarrow \text{Mod}_{\mathcal{O}_Y}$.

Prop. Suppose $\pi : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces. A slight variation on the inverse image allows us to *pull back an $\mathcal{O}_Y$ -module $\mathcal{G}$* to get an $\mathcal{O}_X$ -module $\pi^* \mathcal{G}$.

- $(X, \pi^{-1} \mathcal{O}_Y)$ is a ringed space, and that the morphism of ringed spaces π factors into $(X, \mathcal{O}_X) \rightarrow (X, \pi^{-1} \mathcal{O}_Y) \rightarrow (Y, \mathcal{O}_Y)$.
- $\pi^{-1} \mathcal{G}$ is an $\pi^{-1} \mathcal{O}_Y$ module.
- $\pi^{-1} \mathcal{G} \otimes_{\pi^{-1} \mathcal{O}_Y} \mathcal{O}_X$ is an $\mathcal{O}_X$ -module, which we call $\pi^* \mathcal{G}$ the *pull back* of $\mathcal{G}$
- $\pi^* = \pi^{-1} - \otimes_{\pi^{-1} \mathcal{O}_Y} \mathcal{O}_X$ is a covariant functor $\text{Mod}_{\mathcal{O}_Y} \rightarrow \text{Mod}_{\mathcal{O}_X}$.
- If furthermore $\rho : (W, \mathcal{O}_W) \rightarrow (X, \mathcal{O}_X)$ is another morphism of ringed spaces, there is a natural isomorphism of functors $\rho^* \pi^* \xrightarrow{\sim} (\pi \circ \rho)^*$.
- π^* is left-adjoint to π_* ; more precisely, for any $\mathcal{F} \in \text{Mod}_{\mathcal{O}_X}$, there is a natural bijection $\text{Hom}_{(X, \mathcal{O}_X)}(\pi^* \mathcal{G}, \mathcal{F}) \xrightarrow{\sim} \text{Hom}_{(Y, \mathcal{O}_Y)}(\mathcal{G}, \pi_* \mathcal{F})$.
- Let the following diagram of ringed spaces be commutative:

$$\begin{array}{ccc} W & \xrightarrow{\beta'} & X \\ \downarrow \alpha' & & \downarrow \alpha \\ Y & \xrightarrow{\beta} & Z \end{array}$$

and let $\mathcal{F}$ be an $\mathcal{O}_X$ -module. We define the *push-pull map* to be the composite

$$\beta^*(\alpha_* \mathcal{F}) = \mathcal{O}_Y \otimes_{\beta^{-1} \mathcal{O}_Z} \beta^{-1}(\alpha_* \mathcal{F}) \xrightarrow{\text{id} \otimes \text{bc}} \mathcal{O}_Y \otimes_{\beta^{-1} \mathcal{O}_Z} \alpha'_*(\beta'^{-1} \mathcal{F}) \rightarrow \alpha'_*(\beta'^* \mathcal{F})$$

where $\text{bc} : \beta^{-1}(\alpha_* \mathcal{F}) \rightarrow \alpha'_*((\beta')^{-1} \mathcal{F})$ is the canonical morphism of sheaves of abelian groups.

- Given a morphism of ringed spaces $\pi : X \rightarrow Y$ with $\pi(p) = q$, there is a map of stalks $(\mathcal{O}_Y)_q \rightarrow (\mathcal{O}_X)_p$.
- Given a morphism of rings $\pi^\# : B \rightarrow A$, we define a morphism of ringed spaces $\pi : \text{Spec}(A) \rightarrow \text{Spec}(B)$ as follows: the map of topological spaces is known to us already; we describe the morphism of sheaves $\mathcal{O}_{\text{Spec}(B)} \rightarrow \pi_* \mathcal{O}_{\text{Spec}(A)}$ on $\text{Spec}(B)$ by describing a morphism of sheaves on the distinguished base of $\text{Spec}(B)$. On $D(g) \subset \text{Spec}(B)$, we define

$$\mathcal{O}_{\text{Spec}(B)}(D(g)) \rightarrow \mathcal{O}_{\text{Spec}(A)}(\pi^{-1}D(g)) = \mathcal{O}_{\text{Spec}(A)}(D(\pi^\#g))$$

by $B_g \rightarrow A_{\pi^\#g}$.

Definition. Recall that a *locally ringed space* is a ringed space $(X, \mathcal{O}_X)$ such that the stalks $\mathcal{O}_{X,p}$ are all local rings; a *morphism of locally ringed spaces* $\pi : X \rightarrow Y$ is a morphism of ringed spaces such that the induced map of stalks $\pi^\# : \mathcal{O}_{Y,q} \rightarrow \mathcal{O}_{X,p}$ sends the maximal ideal $m_{Y,q} \rightarrow m_{X,p}$ (a *morphism of local rings*).

This is quite intuitive; if $p \mapsto q$, and g is a function vanishing at q , then it will pull back to a function vanishing at p ; and if g is a function not vanishing at q , then it will pull back to a function not vanishing at p . This is a consequence of the fact that, if $\pi(p) = q$, π includes an inclusion $\kappa(q) \hookrightarrow \kappa(p)$ of residue fields. The morphism of ringed spaces $\pi : \text{Spec}(A) \rightarrow \text{Spec}(B)$ defined by a ring morphism $\pi^\# : B \rightarrow A$ is a morphism of locally ringed spaces. If $\pi : \text{Spec}(A) \rightarrow \text{Spec}(B)$ is a morphism of locally ringed spaces then it is the morphism of locally ringed spaces induced by $\pi^\# : B = \Gamma(\text{Spec}(B), \mathcal{O}_{\text{Spec}(B)}) \rightarrow \Gamma(\text{Spec}(A), \mathcal{O}_{\text{Spec}(A)}) = A$ as we just defined.

Definition. If X and Y are schemes, then a morphism $\pi : X \rightarrow Y$ as locally ringed spaces is called a *morphism of schemes*. The target Y of π is called the *base scheme*.

A morphism of schemes $\pi : X \rightarrow Y$ is a morphism of ringed spaces that looks locally like morphisms of affine schemes; precisely, if $\text{Spec}(A)$ is an affine open subset of X and $\text{Spec}(B)$ is an affine open subset of Y , and $\pi(\text{Spec}(A)) \subset \text{Spec}(B)$, then the induced morphism of ringed spaces is a morphism of affine schemes. In practice, we use the affine cover interpretation and forget completely about locally ringed spaces. In particular, the category of rings and the opposite category of affine schemes are equivalent. Clearly, morphisms of schemes glue. Consequently, morphisms $X \rightarrow \text{Spec}(A)$ are in natural bijection with ring morphisms $A \rightarrow \Gamma(X, \mathcal{O}_X)$. It is one small step to show that some products of schemes exist: if A, B are rings, then $\text{Spec}(A) \times \text{Spec}(B) = \text{Spec}(A \otimes_{\mathbb{Z}} B)$; and if A, B are $\mathbb{C}$ -algebras, then $\text{Spec}(A) \times_{\text{Spec}(\mathbb{C})} \text{Spec}(B) = \text{Spec}(A \otimes_{\mathbb{C}} B)$.

We consider the category of complex schemes (or more generally the category of k -schemes where k is a field, or more generally the category of A -schemes where A is a ring, or more generally the category of S -schemes where S is a scheme); the objects of the latter, Sch_S , called S -schemes, are morphisms of the form

$$\begin{array}{c} X \\ \downarrow \\ S \end{array}$$

The morphism to S is called the *structure morphism*; this is because if $S = \text{Spec}(A)$, the structure morphism gives the functions on each open set of X the structure of an A -algebra. The morphisms

in the category of S -schemes are defined to be commutative diagrams

$$\begin{array}{ccc} X & \longrightarrow & Y \\ \downarrow & & \downarrow \\ S & \xlongequal{\quad\quad\quad} & S \end{array}$$

When there is no confusion (if the base scheme is clear), simply the top row of the diagram is given; in the case where $S = \operatorname{Spec}(A)$ where A is a ring, we get the notion of an A -scheme. A complex geometer may consider the category of $\mathbb{C}$ -schemes. The disjoint union of two schemes is their coproduct in the category of schemes.

Definition. We define the functor of points, and scheme-valued points (and ring-valued points, and field-valued points) of a scheme. If Z is a scheme, then Z -valued points of a scheme X , denoted $X(Z)$, are defined to be maps $Z \rightarrow X$; if A is a ring, then A -valued points of a scheme X , denoted $X(A)$, are defined to be $(\operatorname{Spec}(A))$ -valued points of the scheme.

If you are working over a base scheme B , then in the above definition there is an implicit structure map $Z \rightarrow B$ (or $\operatorname{Spec}(A) \rightarrow B$ in the case of $X(A)$). A morphism of schemes $X \rightarrow Y$ induces a map of Z -valued points $X(Z) \rightarrow Y(Z)$. Note that morphisms of schemes are not determined by their "underlying" map of points; they *are* determined by their induced maps of Z -valued points, as Z varies over all schemes. The notation $X(Z)$ suggests the interpretation of X as a (contravariant) functor h_X from schemes to sets—the *functor of* (scheme-valued) *points* of the scheme X . A "low-brow" reason for A -valued points to be useful: the A -valued points of an affine scheme $\operatorname{Spec} \mathbb{Z}[x_1, \dots, x_n]/(f_1, \dots, f_r)$ (where $f_i \in \mathbb{Z}[x_1, \dots, x_n]$ are relations) are precisely the solutions to the equations $f_1(x_1, \dots, x_n) = \dots = f_r(x_1, \dots, x_n) = 0$ in the ring A .

Suppose that $\phi : S_\bullet \rightarrow R_\bullet$ is a morphism of $(\mathbb{Z}^{\geq 0})$ -graded rings. By map of graded rings, we mean a map of rings that preserves the grading as a map of graded abelian semigroups, or graded commutative monoids; in other words, there is a $d > 0$ such that $S_n \rightarrow R_{dn}$ for all n . Our morphism of graded rings induces a morphism of schemes $(\operatorname{Proj} R_\bullet) \setminus V(\phi(S_+)) \rightarrow \operatorname{Proj} S_\bullet$. In particular, if $V(\phi(S_+)) = \emptyset$, then we have a morphism $\operatorname{Proj} R_\bullet \rightarrow \operatorname{Proj} S_\bullet$. Clearly, if ϕ is furthermore a morphism of A -algebras, then the induced morphism is a morphism of A -schemes.

Suppose $S_\bullet$ is a finitely generated graded ring; we define the n th *Veronese subring* of $S_\bullet$ by $S_{n\bullet} = \bigoplus_{j=0}^\infty S_{nj}$. The "old degree" n is the "new degree" 1. The map of graded rings $S_{n\bullet} \hookrightarrow S_\bullet$ induces an isomorphism $\operatorname{Proj} S_\bullet \xrightarrow{\sim} \operatorname{Proj} S_{n\bullet}$. Informally, a rational map is a morphism defined almost everywhere. *Take X to be reduced in the discussion of rational maps that follows.*

Definition. A *rational map* $\pi : X \dashrightarrow Y$ of reduced schemes is the data of a morphism $\alpha : U \rightarrow Y$ from a dense open set $U \subset X$, with the equivalence relation $(\alpha : U \rightarrow Y) \sim (\beta : V \rightarrow Y)$ if there is a dense open set $Z \subset U \cap V$ such that $\alpha|_Z = \beta|_Z$. Note that a rational map need not be a map. The rational map of S -schemes for a scheme S is defined the same way, except X, Y are S -schemes and $\alpha : U \rightarrow Y$ is a morphism of S -schemes.

Example. A morphism is a rational map. Another important example is the projection $\mathbb{P}_A^n \dashrightarrow \mathbb{P}_A^{n-1}$ given by $[x_0, \dots, x_n] \mapsto [x_0, \dots, x_{n-1}]$.

A rational map $\pi : X \dashrightarrow Y$ is *dominant* if for some (and hence every) representative $U \rightarrow Y$, the image is dense in Y ; a morphism is a *dominant morphism* if it is dominant as a rational map. A rational map of irreducible schemes is dominant iff π sends the generic point of X to the generic point of Y .

Definition. A rational map $\pi : X \dashrightarrow Y$ is said to be *birational* if it is dominant, and there is another rational map (a "rational inverse") ψ that is also dominant, such that $\pi \circ \psi$ is (in the same equivalence class as) the identity on Y , and $\psi \circ \pi$ is (in the same equivalence class as) the identity on X . A morphism is a *birational morphism* if it is birational as a rational map. We say that X, Y are birational to each other if there exists a birational map $X \dashrightarrow Y$.

If X, Y are irreducible, then birational maps induced isomorphism of function fields. An integral finite type k -scheme is said to be rational if it is birational to $\mathbb{A}_k^n$ for some n .

- Suppose X and Y are reduced schemes. Then X and Y are birational iff there is a dense open subscheme U of X and a dense open subscheme V of Y such that $U \cong V$.
- Suppose X is an integral k -scheme and Y is an integral finite type k -scheme, and we are given an extension of function fields $\phi^\# : K(Y) \hookrightarrow K(X)$ preserving k . Then there exists a dominant rational map of k -schemes $\phi : X \dashrightarrow Y$ inducing $\phi^\#$.

If X is a scheme, there is a bijection between the maps $X \rightarrow \mathbb{A}^1$ and global sections of the structure sheaf. Maps $\pi : X \rightarrow \mathbb{A}_{\mathbb{Z}}^1$ correspond to maps of rings $\pi^\# : \mathbb{Z}[t] \rightarrow \Gamma(X, \mathcal{O}_X)$ and $\pi^\#(t)$ is a function on X ; this is reversible. We can restate this as follows. We have two different contravariant functors from Sch to $Sets$: maps to $\mathbb{A}^1$ (i.e., $H : X \mapsto Mor(X, \mathbb{A}_{\mathbb{Z}}^1)$) and functions on X ($F : X \mapsto \Gamma(X, \mathcal{O}_X)$).